

LearnIQ

TRIGONOMETRY LEARNING GUIDE

Learn at your level. Learn your way.

1. What is Trigonometry?

Trigonometry is the study of the relationship between the sides and angles of triangles. It is widely used in engineering, construction, architecture, navigation and science.

Main idea: If we know some sides or angles of a right triangle, trigonometry can help us find the unknown parts.

2. Parts of a Right Triangle

Side	Meaning
Hypotenuse	Side opposite the 90° angle
Opposite	Side opposite the angle θ
Adjacent	Side next to θ , excluding the hypotenuse

Tip: First find the angle you are working with. Then identify the opposite, adjacent and hypotenuse sides.

3. The Three Trigonometric Ratios

Ratio	Formula	Sides Used
$\sin \theta$	Opposite / Hypotenuse	O and H
$\cos \theta$	Adjacent / Hypotenuse	A and H
$\tan \theta$	Opposite / Adjacent	O and A

Easy memory trick: SOH - CAH - TOA

SOH $\rightarrow$ Sin = Opposite / Hypotenuse

CAH $\rightarrow$ Cos = Adjacent / Hypotenuse

TOA $\rightarrow$ Tan = Opposite / Adjacent

4. Important Trigonometric Values

Angle	$\sin \theta$	$\cos \theta$	$\tan \theta$
0°	0	1	0
30°	$1/2$	$\sqrt{3}/2$	$1/\sqrt{3}$
45°	$1/\sqrt{2}$	$1/\sqrt{2}$	1
60°	$\sqrt{3}/2$	$1/2$	$\sqrt{3}$

90°	1	0	Undefined
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5. Worked Example

Question: A right triangle has an angle of 30° and a hypotenuse of 10 cm. Find the opposite side.

Step 1: We know the opposite side and hypotenuse are involved, so use sine.

Step 2: $\sin 30^\circ = \text{Opposite} / \text{Hypotenuse}$

Step 3: $1/2 = \text{Opposite} / 10$

Step 4: Opposite = 5 cm

Answer: 5 cm

6. Trigonometry in the Real World

Imagine standing some distance away from a tall building. You can measure the distance to the building and the angle from the ground to the top. Trigonometry can then help estimate the building's height.

Trigonometry is used in surveying, construction, robotics, navigation, engineering and astronomy.

7. Quick Practice

1. What is $\sin 30^\circ$?
2. What is $\cos 60^\circ$?
3. What is $\tan 45^\circ$?
4. Which ratio uses opposite and adjacent sides?
5. Which side is opposite the 90° angle?

Answers:

1. $1/2$
2. $1/2$
3. 1
4. $\tan \theta$
5. Hypotenuse